

DEGREE CERTIFICATE

The Vice-Chancellor of Université de Lorraine hereby certifies that
the following degree:

Master de Sciences, Technologies, Santé mention Informatique Parcours type
Informatique Décisionnelle (I&D),

was awarded to :

AHMED TIOUAL

born on 02/27/2003 in GUELMIM

following successful completion of the academic year 2025 / 2026 with the
honourable mention Bien

This degree is constituted of 120 European credits

Nancy, the 09/19/2026



La Présidente, Hélène BOULANGER

National student identifier : 243380208GH

Student number : 32525387



3 Courses

**AWS Cloud Technical
Essentials**

Migrating to the AWS Cloud

**Architecting Solutions on
AWS**



Jul 14, 2024

AHMED TIOUAL

has successfully completed the online, non-credit Specialization

AWS Fundamentals

In this Specialization, learners gained proficiency in essential concepts, services, and use cases within the Amazon Web Services (AWS) cloud ecosystem, including core AWS services and key AWS security concepts. The Specialization also covered fundamental strategies for planning and migrating existing workloads to AWS and how to build and deploy serverless applications with AWS. Learners are given opportunities to solidify their understanding by engaging in various hands-on labs and exercises throughout the Specialization.

AWS Fundamentals

The online specialization named in this certificate may draw on material from courses taught on-campus, but the included courses are not equivalent to on-campus courses. Participation in this online specialization does not constitute enrollment at this university. This certificate does not confer a University grade, course credit or degree, and it does not verify the identity of the learner.

Verify this certificate at:

<https://coursera.org/verify/specialization/BYVW8W4AUTTP>



Apr 18, 2024

AHMED TIOUAL

has successfully completed

ETL and Data Pipelines with Shell, Airflow and
Kafka

an online non-credit course authorized by IBM and offered through Coursera

Yan Luo Ramesh Sannareddy S. S. M. Jeff Grossman

Yan Luo, Ramesh Sannareddy
Sabrina Spillner, Jeff Grossman

COURSE
CERTIFICATE



Verify at:
<https://coursera.org/verify/PRZ7CATQRWX6>

Coursera has confirmed the identity of this individual and
their participation in the course.



May 14, 2024

AHMED TIOUAL

has successfully completed

Generative AI: Prompt Engineering Basics

an online non-credit course authorized by IBM and offered through Coursera

Antonio Cangiano

Antonio Cangiano
Engineering Manager and AI Specialist
IBM Digital Business Group

Rav Ahuja

Rav Ahuja
Global Program Director,
Skills Network

COURSE
CERTIFICATE



Verify at:
<https://coursera.org/verify/VHWU4ANMBHDX>

Coursera has confirmed the identity of this individual and
their participation in the course.



3 Courses

Supervised Machine Learning: Regression and Classification

Advanced Learning Algorithms

Unsupervised Learning, Recommenders, Reinforcement Learning



Aug 4, 2024

AHMED TIOUAL

has successfully completed the online, non-credit Specialization

Machine Learning

Congratulations on completing all three courses of the Machine Learning Specialization! You studied modern machine learning concepts, including supervised learning (linear regression, logistic regression, neural networks, decision trees), unsupervised learning (clustering, anomaly detection), recommender systems, and reinforcement learning. You learned some of the best practices for building machine learning models. You've also gained practical skills to apply machine learning techniques to challenging real-world problems. Now #BreakIntoAI and start building your career in machine learning!

Andrew Ng, Instructor,
DeepLearning.AI

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Verify this certificate at:

<https://coursera.org/verify/specialization/S0IIS7LK9X3M>